

QC Flag Diagnostics: Control-Probe Failures by Cohort

Which ewastools control metrics drive control-only sample flags

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Contents

1	ROSMAP	1
2	ROSMAP_APOE4	2
3	MSBB	2
4	MOA-PAD	3

1 ROSMAP

```
## flag_reason distribution: ## ## none control bisulfite;control detP;control
## 645 78 9 7 ## detP ## 4 ## ## 78 control-only samples
```

```
## ## ## failed metric per control-only sample ## sample failed_metric ##
5822062002_R06C01 Non.polymorphic.Green ## 5815381005_R02C01
Non.polymorphic.Green ## 5822038004_R03C01 Non.polymorphic.Green ##
5822071004_R02C02 Non.polymorphic.Green ## 5822071007_R06C01
Non.polymorphic.Green ## 5815381023_R01C02 Non.polymorphic.Green ##
5822038009_R03C02 Non.polymorphic.Green ## 5822038004_R05C01
Non.polymorphic.Green ## 5822054008_R01C01 Non.polymorphic.Green ##
5822038004_R04C01 Non.polymorphic.Green ## 5815381025_R04C02
Non.polymorphic.Green ## 5822062002_R02C02 Non.polymorphic.Green ##
5815381023_R05C01 Non.polymorphic.Green ## 5815381006_R02C02
Non.polymorphic.Green ## 5815381025_R02C01 Non.polymorphic.Green ##
5815381009_R05C01 Non.polymorphic.Green ## 5822038004_R01C01
Non.polymorphic.Green ## 5815381029_R01C01 Non.polymorphic.Green ##
5815381014_R01C01 Non.polymorphic.Green ## 5822038009_R02C02
Non.polymorphic.Green ## 5815381018_R06C01 Non.polymorphic.Green ##
5822071001_R04C01 Non.polymorphic.Green ## 5815381002_R01C01
Non.polymorphic.Green ## 5822038009_R02C01 Non.polymorphic.Green ##
5815381005_R01C02 Non.polymorphic.Green ## 5815381018_R01C01
Non.polymorphic.Green ## 5822038007_R04C01 Non.polymorphic.Green ##
5815381009_R03C02 Non.polymorphic.Green ## 5815381006_R04C01
Non.polymorphic.Green ## 5815381025_R06C01 Non.polymorphic.Green ##
5822054008_R02C01 Non.polymorphic.Green ## 5815381026_R02C01
Non.polymorphic.Green ## 5815381017_R06C02 Non.polymorphic.Green ##
5822071006_R02C01 Non.polymorphic.Green ## 5822038006_R02C01
```

```

Non.polymorphic.Green ## 5822054002_R05C01 Non.polymorphic.Green ##
5815381005_R06C02 Non.polymorphic.Green ## 5815381017_R02C02
Non.polymorphic.Green ## 5822071001_R05C01 Non.polymorphic.Green ##
5822038001_R01C01 Non.polymorphic.Green ## 5815381025_R01C02
Non.polymorphic.Green ## 5815381025_R04C01 Non.polymorphic.Green ##
5822071004_R01C02 Non.polymorphic.Green ## 5822071001_R06C02
Non.polymorphic.Green ## 5822062002_R05C01 Non.polymorphic.Green ##
5822071001_R04C02 Non.polymorphic.Green ## 5822038004_R06C01
Non.polymorphic.Green ## 5815381025_R05C01 Non.polymorphic.Green ##
5815381002_R01C02 Non.polymorphic.Green ## 5815381029_R03C01
Non.polymorphic.Green ## 5815381008_R03C01 Non.polymorphic.Green ##
5822038009_R01C01 Non.polymorphic.Green ## 5815381023_R06C01
Non.polymorphic.Green ## 5822038006_R03C02 Non.polymorphic.Green ##
5822038007_R01C01 Non.polymorphic.Green ## 5815381006_R06C01
Non.polymorphic.Green ## 5815381002_R06C02 Non.polymorphic.Green ##
5822038009_R04C02 Non.polymorphic.Green ## 5822038004_R05C02
Non.polymorphic.Green ## 5815381002_R02C01 Non.polymorphic.Green ##
5822062002_R06C02 Non.polymorphic.Green ## 5822038001_R06C01
Non.polymorphic.Green ## 5822038009_R01C02 Non.polymorphic.Green ##
5815381018_R02C01 Non.polymorphic.Green ## 5815381023_R05C02
Non.polymorphic.Green ## 5815381026_R02C02 Non.polymorphic.Green ##
5822038009_R05C02 Non.polymorphic.Green ## 5815381029_R06C01
Non.polymorphic.Green ## 5815381019_R06C01 Non.polymorphic.Green ##
5815381029_R04C02 Non.polymorphic.Green ## 5815381029_R04C01
Non.polymorphic.Green ## 5815381002_R06C01 Non.polymorphic.Green ##
5815381014_R01C02 Non.polymorphic.Green ## 5822038008_R05C02
Non.polymorphic.Green ## 5822071001_R05C02 Non.polymorphic.Green ##
5815381025_R01C01 Non.polymorphic.Green ## 5815381027_R02C02
Non.polymorphic.Green ## 5815381002_R05C01 Non.polymorphic.Green ## ## ## tally
of metrics driving control-only drops ## Non.polymorphic.Green ## 78 ## ## ##
summary(Non.polymorphic.Green) among control-only samples ## Min. 1st Qu.
Median Mean 3rd Qu. Max. ## 3.947 4.577 4.742 4.687 4.912 4.998 ## ## ## top
sample-prefix counts among control-only samples ## ## 5815381025 5822038009
5815381002 5822038004 5815381029 5822071001 5815381023 ## 7 7 6 6 5 5 4 ##
5822062002 5815381005 5815381006 ## 4 3 3

```

All flags (n=78) were driven by Non-polymorphic Green marginally below the Illumina cutoff (range 3.95–5.0) with no other QC failure, consistent with a systematic control-probe artifact rather than sample failure.

2 ROSMAP_APOE4

```

## flag_reason distribution: ## ## none bisulfite;control ## 1015 10 ##
beadcount detP;sex;bisulfite;control ## 1 1 ##
detP;sex;bisulfite;control;beadcount ## 1 ## ## 0 control-only samples

```

```

## [ROSMAP_APOE4] no samples with flag_reason == 'control' exactly (control
failures may only occur combined with other flags here)

```

3 MSBB

```

## flag_reason distribution: ## ## none bisulfite;control ## 188 13 ## ## 0

```

control-only samples

[MSBB] no samples with flag_reason == 'control' exactly (control failures
may only occur combined with other flags here)

4 MOA-PAD

flag_reason distribution: ## ## none bisulfite;control ## 220 1 ##
detP;intensity;sex;bisulfite;control sex ## 1 1 ## ## 0 control-only samples

[MOA-PAD] no samples with flag_reason == 'control' exactly (control failures
may only occur combined with other flags here)